
Visa type H4-EAD (Authorized to work in the US/ No Sponsorship required)

I am a data-driven researcher with an expertise in **Python object-oriented programming, data engineering, and large-scale data analysis**. 7+ years of experience working with complex, high-dimensional datasets, simulation pipelines, and statistical modeling. Strong track record of analyzing and delivering insights from multi-terabyte data, building reproducible workflows, and collaborating with international teams. I am currently working as a full-time postdoctoral researcher at NYU Abu Dhabi.

SKILLS

Programming & Tools	Python-OOP (NumPy, Pandas, SciPy, Matplotlib, scikit-learn, JAX), Git, Jupyter, $\LaTeX$, VSCode, PyPI, Sphinx
Quantitative tools	Simulation data processing, Data pipelines, HPC environments, Statistical modeling, Data visualization
Communication	Marathi (Mother tongue), Hindi (Fluent), English (Fluent), German (B2)

RESEARCH EXPERIENCE

CodeAstro Project <i>CIERA, Northwestern University, Chicago, IL</i>	Aug. 4 – 8, 2025 <i>Workshop</i>
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Skills: Python, NumPy, Pandas, JAX, SciPy, Matplotlib, PyPI, Sphinx, GitHub, VSCode
Remotely collaborated with a team and developed an end-to-end open-source Python package (with complete documentation) that classifies galaxies from numerical simulations morphologically

HELLO-SKIRT Catalog: Dust post-processing in high-z simulated regime (in prep.) <i>New York University Abu Dhabi, UAE</i>	Dec. 2023 – Present <i>Postdoc project</i>
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Skills: Python, NumPy, Matplotlib, $\LaTeX$, GitHub, Jupyter
Analysed large simulation datasets over cloud server and developed a python pipeline that simulates the effects of cosmic dust on galaxies

Cold gas & its origin in satellite galaxies in TNG50 simulations <i>Centre for Computational Astrophysics (USA), University of Potsdam (Germany)</i>	Nov. 2022 – Nov. 2024 <i>Ph.D. project</i>
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Skills: Python, NumPy, Matplotlib, $\LaTeX$, GitHub, Jupyter
Analysed large datasets from TNG simulations over online Jupyter notebook to find correlations between cold gas and satellite galaxy properties and published results in peer-reviewed journal

Gas distribution in the HESTIA Local Group simulations <i>University of Potsdam (Germany)</i>	2019 – 2021 <i>Ph.D. project</i>
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Skills: Python, NumPy, Matplotlib, $\LaTeX$, GitHub, Jupyter
Analysed simulation data to study the gas distribution around Milky Way & Andromeda galaxies

EDUCATION

Postdoctoral researcher , <i>New York University Abu Dhabi</i>	Oct. 2023 – Present
Ph.D. Astrophysics , <i>University of Potsdam (Supervisor: Dr. Martin Sparre) (Magna cum laude)</i>	Oct. 2019 – March 2023
M.Sc. Astrophysics , <i>University of Potsdam (Distinction)</i>	Sept. 2019

ACTIVITIES

Conference talk , Santa Cruz Galaxy Workshop, UCSC	2025
Dissertation talk , 244th American Astronomical Society Meeting, Madison	2025
Computer Simulations , Public Talk, The Sheikh Zayed Private Academy for Girls, Abu Dhabi, UAE	2024
TA (Astrophysics) , University of Potsdam, Germany	2021 – 2022
Astronomy vs. Astrology , Public Talk (online) in Marathi , Breakthrough Science Society webinar series	July 2021
Numerical Simulations & our Universe , Public Talk, AstroZoom webinar series	Jan. 2021
Voluntary astronomy teaching for school students vidyaloka	2020 – 2021
Member , CARINAS , Supernova Foundation & PENTA	2020-Present
TA (Physics) , Mumbai	2013 – 2016